1. Clinical Question & Core Hypothesis

Clinical Question

EMT/Wnt-driven metastatic programs are linked to metabolic reprogramming (GLS) and cell-cycle activation (cyclin D1), yet tumors with apparent immune infiltration may still fail immunotherapy — why does immune presence not translate into therapeutic benefit in these metabolically/proliferatively active tumors? 2) Tier 1 paths used: the single path highlights GLS (glutaminase) and cyclin D1 (CCND1) as mechanistic anchors connecting metastatic/EMT programs to metabolic support and proliferative drive; I built the core hypothesis on GLS and CCND1 because they directly provide a plausible mechanism (metabolic suppression + rapid cell-cycle) that can blunt T-cell function. 3) Tier 2 supporting evidence: CXCR4 (chemokine receptor) and the upstream Wnt/EMT axis provide context for recruitment, trafficking and induction of the GLS–CCND1 program that could co-occur with immune infiltration; these reinforce the mechanism but are not central to the focused claim. 4) Choice of endpoint: I selected therapy_response (immunotherapy sensitivity/resistance) because the path explicitly connects GLS activity to immunotherapy outcomes, making a mechanistic, clinically actionable question. 5) Deliberate exclusions: I excluded EMT/Wnt as core variables (kept them as secondary context) to respect the 1–2 anchor gene limit and preserve mechanistic clarity; I also excluded chemotherapy agents and transporter proteins since they dilute the immunotherapy-focused question and derive from more distal or correlative edges in the path?

Working Hypothesis

TIL-high tumors with elevated GLS-driven glutaminolysis and CCND1-driven proliferation exhibit impaired antitumor immunity due to metabolic competition and rapid cell-cycle progression that suppress T-cell effector function. Therefore, high TIL density will not predict favorable immunotherapy response in GLS-high/CCND1-high tumors, and concurrent targeting of GLS and/or CCND1 pathways will be required to restore therapeutic sensitivity.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	GLS, CCND1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GLS: GLS) and cell-cycle activation (cyclin D1), yet tumors with apparent immune infiltration may still fail immunotherapy — why does immune presence not translate into therapeutic benefit in these metabolically/proliferatively active tumors? 2) Tier 1 paths used: the single path highlights GLS (glutaminase) and cyclin D1 (CCND1) as mechanistic anchors connecting metastatic/EMT programs to metabolic support and proliferative drive; I built the core hypothesis on GLS and CCND1 because they directly provide a plausible mechanism (metabolic suppression + rapid cell-cycle) that can blunt T-cell function.

CCND1: CCND1) as mechanistic anchors connecting metastatic/EMT programs to metabolic support and proliferative drive; I built the core hypothesis on GLS and CCND1 because they directly provide a plausible mechanism (metabolic suppression + rapid cell-cycle) that can blunt T-cell function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CXCR4	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: analyze public ICB datasets for co-expression of GLS and CCND1 with TIL density and response;

Prediction 2: perform multiplex IHC or spatial transcriptomics to map GLS/CCND1 expression with TILs in pretreatment biopsies;

Prediction 3: in vitro: co-culture human tumor cells with T cells under varying GLS activity (CB-839) and cell-cycle inhibition (CDK4/6 inhibitors) to assess T-cell cytotoxicity and cytokine production;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CXCR4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.